Methods and tools from the kickoff

Every method, metric and tool named in the kickoff slides or spoken in the session, explained in one or two lines, with a link. "Slide" is the kickoff slide number | Team 1 | 11 October 2026

1. Workflow concepts

Concept	Slide	What it means	Why it matters here	How / link
Train / test split	43, 45	Hold back part of the patients. The model learns from the training set ("practice problems") and is checked once on the test set ("the final exam")	Without it, accuracy is measured on patients the model has already seen and looks better than it is	Base R `sample()`; rsample · scikit-learn
Stratified split	45	Split so that training and test sets have the same mix of outcomes (and here, onset phenotype)	The organizers' example of a biased split: all adult onset in training, juvenile only in test	Split within each outcome group; bin a continuous outcome first
Fixed random seed	45	A number that makes "random" choices repeatable	The split and results can be reproduced by teammates and judges	`set.seed()` · `random_state=`
Leakage	46	Information from the test set sneaking into model building	Inflates scores. Rule: split first, fit on train only, apply to test	scikit-learn cross-validation guide
Preprocessing	46	Cleaning and transformations a method needs: encoding categories as numbers, scaling numeric columns	Anything learned from data (means, SDs) must come from the training set only	`model.matrix()`, `scale()` · scikit-learn preprocessing
Cross-validation	43, 56	Repeatedly hold out a slice of the training set to compare settings, without touching the test set	Needed to choose the LASSO penalty and to tune ML models	`cv.glmnet()`, `xgb.cv()` · scikit-learn
Hyperparameter tuning (grid search)	71	Every model has settings, such as tree depth. Grid search tries combinations and keeps the best ("oven temperature and time")	"A large grid on a small arm mostly finds noise. Keep grids coarse." Show untuned and tuned	Loop over a small grid · GridSearchCV
Overfitting	58, 65, 69	A model that memorises noise in the training data and predicts new patients poorly	The main risk with about 125 training patients per arm; favours simple or penalised models	ISLR, chapters 2 and 5
Parsimony	55 (spoken)	Prefer the simplest model that performs at the same level	Stated as a principle: "always there"	BIC instead of R^2 when selecting

2. Covariate (statistical) models

Method	Slide	What it does	For our data / watch out	How / link
Linear regression	53	A straight line between each covariate and a continuous outcome; the slope is the effect size	Natural first step. Spoken: "in many cases actually outperforms the more fancy methods". Assumes straight-line effects; sensitive to outliers	`lm()` · R documentation
Logistic regression	64	A weighted scorecard: each covariate adds or removes points and the total becomes a probability of response	Transparent baseline for the responder label. Misses interactions unless added	`glm(family = binomial)` · R · scikit-learn
Stepwise selection	54	Adds (forward) or removes (backward) covariates one at a time, or both, judged by AIC or BIC ("packing a suitcase")	Unstable: with correlated covariates the order decides which stays. "Don't treat the final model as the truth"	`MASS::stepAIC()` · MASS
Best subset selection	55	Fits every combination of covariates and keeps the best of each size	Feasible with 18 covariates. Use BIC, not R^2, which never penalises extra covariates	`leaps::regsubsets()` · leaps

LASSO	56	Regression with a penalty that pushes weak coefficients to exactly zero, leaving a short list ("a budget constraint")	Selects and predicts. May arbitrarily drop one of two correlated covariates	`cv.glmnet(alpha = 1)` · glmnet vignette
Ridge	56	Penalty that shrinks all coefficients toward zero but never to zero	Stabilises many correlated covariates; does not select	`cv.glmnet(alpha = 0)` · glmnet
Elastic net	56	A mix of LASSO and ridge: sets some coefficients to zero and shrinks correlated ones together	A good default when covariates overlap (age, onset, duration)	`cv.glmnet(alpha = 0.5)` · glmnet
Group / adaptive LASSO, SCAD, MCP	56	Variants of the penalty, named on the slide as "add more"	Not needed for a first answer	ISLR, chapter 6
Polynomial regression	57	Adds squared or cubed terms so the line can bend	High degrees swing wildly at the edges of the data (Runge's phenomenon)	`lm(y ~ poly(x, 2))` · ISLR ch. 7
Piecewise / step model	57	Different straight lines or levels in different ranges of a covariate	Breakpoints need careful choice	`cut()` then `lm()` · ISLR ch. 7
Regression spline	58	Polynomial pieces joined smoothly at chosen points ("knots")	More knots, more flexibility, more overfitting	`splines::bs()` · splines
Smoothing spline	58	A spline that sets its own smoothness through a penalty	No knot choice needed, but harder to interpret	`smooth.spline()` · splines
Generalised additive model (GAM)	59	Regression where each covariate may have its own smooth curve, all added together	Handles both outcome types and gives one effect plot per covariate. Smoothness setting k controls over- or under-fitting	`mgcv::gam()` · mgcv · Noam Ross tutorial
Principal component analysis (PCA)	60	Combines correlated numeric covariates into a few uncorrelated summary components	Useful to see structure. Components are blends, so hard to explain or stratify on; ignores the outcome	`prcomp(scale. = TRUE)` · R · StatQuest video
MCA and FAMD	spoken	Relatives of PCA: multiple correspondence analysis for categorical data, factor analysis of mixed data for both types	Same interpretability drawback as PCA	FactoMineR

3. Machine-learning models

Method	Slide	What it does	For our data / watch out	How / link
Decision tree	65	A chain of yes/no questions on covariates, like a triage flowchart; each path ends in a prediction	Gives cut-offs that can be read aloud. One deep tree memorises noise, so limit the depth	`rpart::rpart()` · rpart · scikit-learn
Random forest	66	Many trees grown on random slices of the data, which then vote ("a panel of clinicians")	Captures interactions without specifying them and ranks covariates. Harder to read than one tree	`ranger::ranger()` · ranger · scikit-learn
XGBoost	69	Small trees built one after another, each fixing the errors of the last ("a student who reviews only the questions they got wrong")	Often strong on tabular clinical data. Many settings; overfits quickly with few patients	`xgboost::xgb.train()` · R package · tutorial
Support vector machine	67	Finds the boundary that separates two groups with the widest gap; can be curved	Gives a boundary, not rules; hard to explain to clinicians	e1071 · scikit-learn
K-nearest neighbours	68	Finds the K most similar past patients and copies their majority outcome ("patients like you")	Intuitive, but noise covariates dilute similarity. Spoken: "really sensitive" to this	class · scikit-learn
Neural network (MLP)	70	A hidden layer blends covariates into new features; an output layer predicts	Use a deliberately tiny one if at all. The least transparent model; small arms limit what it can learn	nnet · scikit-learn
Choosing a model	74	A flowchart for picking an estimator from data size and task	Orientation only	scikit-learn map

4. Seeing which covariates a model used

Tool	Slide	What it does	For our data / watch out	How / link
Coefficients and selected terms	81	For covariate models: which terms stepwise kept, and which coefficients LASSO or elastic net left non-zero	The direct answer to "which covariates matter, and in which direction"	`coef()`
Permutation importance	72	Shuffle one covariate; if predictions get worse, the model relied on it ("removing one ingredient from a recipe")	Ranks covariates per arm. Correlated covariates share the credit	`ranger(importance = "permutation")` · vip · scikit-learn
SHAP	73	Splits one patient's prediction into a contribution from each covariate, with direction ("an itemised receipt")	"It explains the model, not biology. Importance does not prove causation"	shapviz (R) · shap (Python)
Covariate-by-method grid	81	A table of which method picked which covariate	"Agreement across methods builds confidence. A covariate only one method picks needs a second look"	Built from the above
Interpretation book	74	Free book on interpreting ML models	Background reading	Interpretable Machine Learning

Our script adds three noise checks that were not taught at kickoff, all logged in `DECISIONS.md`: a random "probe" column as a negative control, LASSO refitted on 200 resamples to see how often each covariate is kept, and Boruta, which tests whether a covariate beats shuffled copies of the covariates in a random forest. Whether to keep Boruta or use plain permutation importance is an open team decision.

5. Metrics

Metric	Slide	What it measures	Reading it
RMSE	76	Typical size of the prediction error for a continuous outcome, penalising large misses more	Lower is better. In the outcome's units, so not comparable across arms
MAE	76	Average size of the prediction error	Lower is better. Same units caveat
R^2	76	Share of variation explained, compared with always predicting the test-set average	1 perfect; 0 no better than the average; below 0 worse than the average
Confusion matrix	77	The four outcomes of a yes/no prediction: responder found, responder missed, false alarm, non-responder found	Every classification metric is built from these counts
Recall	77	Of the true responders, how many did we find?	Report for the responder class
Precision	77	Of the predicted responders, how many were right?	Report with recall
Accuracy	77–78	Share of all patients classified correctly	Misleads when responders are the minority; never report alone. State the cut-off used
ROC-AUC	79	Whether the model ranks true responders above non-responders, across every cut-off	0.5 is guessing, 1.0 perfect. Ignores class mix. Not the pharmacokinetic AUC
Bootstrap interval	80	Resample the test patients many times to show how uncertain a score is	"With only a few dozen test patients per arm, one patient moves the scores noticeably"
AIC, BIC	54–55	Fit criteria that reward good fit and penalise extra covariates; BIC penalises more	Used to choose among models during selection

Reference for all classification and regression metrics: scikit-learn model evaluation. In R, ROC-AUC is available in pROC; our script computes it directly.

6. Software and resources

Tool	Role	Link
R	Used for the kickoff's covariate-model examples; our team's main language	cran.r-project.org
Python, scikit-learn	Used for the kickoff's ML examples. Any language is allowed	scikit-learn.org

ISLR textbook	"An Introduction to Statistical Learning", free, with R and Python editions. The kickoff's main reference	statlearning.com
Code-first tutorials	Blog listed on the resources slide	machinelearningmastery.com
caret	An R framework that wraps many of the above behind one interface (our addition; not named at kickoff)	caret
AI tools	LLM chatbots and AI agents are welcome. Report which tools and for what, in code comments and in the talk; understand every step; the team is accountable	Slide 48
SharePoint	Team folder for sharing files, posting questions to the organizers and submitting slides and code	Team 1 folder

7. Clinical measures named in the session

Measure	What it is	Where it appears
Myotonia severity	How slowly muscle relaxes after contraction	Baseline covariate; its 12-week change is the mexiletine outcome (negative is better)
QMT	Quantitative Muscle Testing, a measured strength composite	Its 26-week change is the oligonucleotide outcome (positive is better)
vHOT	Video hand opening time: seconds to open the hand after a grip	Common trial endpoint for myotonia; not in the dataset because it varies widely
Combined cognitive + motor score	Improvement score from 0 to 20	The tideglusib outcome at 26 weeks (higher is better)
Six-minute walk distance	Distance walked in six minutes	Mentioned only: unchanged by mexiletine in a six-month trial
CDM1-RS, CGI-I	Clinician rating scale for congenital DM1; Clinical Global Impression of Improvement	Mentioned only: endpoints of the REACH-CDM tideglusib trial
PR interval, CK	ECG conduction time; creatine kinase, a blood marker of muscle damage	Baseline covariates

Compiled from the ACoP 2026 Hackathon kickoff deck and session notes. One-line explanations paraphrase the slides; quoted phrases are the organizers'. Links checked 11 October 2026. Prepared with Claude (Anthropic).